

This section explains **what the Fraud Management System (FMS) should do**. Think of it as an intelligent security system that continuously watches insurance transactions, predicts fraud, and helps investigators act quickly.

1. Ingest and prepare legacy data for model training

Simple Meaning

The system should collect **old historical data** (legacy data) from existing insurance systems and clean it before using it to train AI models.

Why?

AI cannot detect fraud unless it learns from previous cases.

Example:

Old claim records

Customer details

Payment history

Policy information

Past fraud cases

Investigation reports

All this data is cleaned and converted into a format AI understands.

Example

Suppose OICL has

- 10 million claim records
- 20,000 confirmed fraud cases

The AI studies these cases and learns what fraud usually looks like.

2. Develop supervised and unsupervised AI models

This means the system should use different AI techniques to identify fraud.

There are two major approaches.

A. Supervised Learning

AI learns using **already labeled data**.

Meaning:

The historical data already tells AI

✓ Fraud

✓ Genuine

Example

Claim	Label
Fake Hospital Bill	Fraud
Genuine Accident	Genuine
Duplicate Claim	Fraud
Normal Claim	Genuine



AI learns these patterns.

Later when a new claim arrives, AI predicts whether it is fraud.

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B. Unsupervised Learning

Sometimes nobody knows which claims are fraud.

AI finds unusual behavior by itself.

Example

Normally

Claim amount = ₹20,000

Suddenly

One customer submits

₹18 lakh claim

AI notices

"This is very unusual."

Even if nobody labeled it as fraud.

Pattern Recognition

Pattern Recognition means

Finding repeated suspicious behaviour.

Example

Every fraudster

Uses same mobile number

Uses same hospital

Uses same bank account

Claims every 15 days

AI connects these patterns.

Machine Learning

Machine Learning improves automatically from data.

Example

If investigators confirm

"This was fraud"

AI remembers it.

Next time

It catches similar fraud faster.

Deep Learning

Deep Learning is a more advanced AI.

It handles

Very large datasets

Complex relationships

Images

Documents

Voice

Scanned claim forms

Medical reports

Signatures

3. Compute fraud probability and risk scores

The AI should calculate how risky each claim or policy is.

Instead of simply saying

Fraud

or

Not Fraud

It gives a score.

Example

Claim	Fraud Score
Claim A	10%
Claim B	55%
Claim C	95%

Higher score

Higher risk

Underwriting Risk Score

Before selling insurance

AI checks

Customer history

Previous claims

Address

Occupation

Age

Medical history

Previous fraud

Then gives a risk score.

Example

Customer A

Risk = 12%

Customer B

Risk = 85%

Claim Risk Score

Whenever a claim arrives

AI calculates

Probability of fraud.

Loop Back Mechanism

Very important requirement.

After investigation

Suppose AI predicted

Fraud

But investigators found

Genuine

The final decision goes back into AI.

AI learns from mistakes.

This is called

Feedback Loop

or

Loop Back Mechanism.

4. Continuously retrain models

Fraudsters keep changing tricks.

Therefore AI should also keep learning.

Every day

Every week

Every month

New fraud data should retrain AI.

Example

Old fraud

Fake hospital bills

New fraud

AI-generated fake documents

The model learns the new fraud pattern.

5. Generate reports and dashboards

The system should create visual reports.

For

Management

Fraud Team

Government

Regulators

FIU

Dashboard may show

Today's fraud attempts

High-risk claims

Money saved

Top fraud locations

Most suspicious hospitals

Most suspicious agents

Monthly fraud trend

6. Reports for FIU and regulators

FIU means

Financial Intelligence Unit (FIU-India)

Insurance companies must report suspicious financial activities.

The system should generate reports automatically.

7. Exception reports

Exception means

Anything unusual.

Example

Normally

Claims are under ₹50,000.

Suddenly

One branch approves

₹75 lakh claim.

This becomes an exception report.

Another example

Same Aadhaar used in 20 policies.

Exception report generated.

8. Escalation Matrix

Whenever fraud is detected

The system should automatically notify the correct people.

Example

AI detects fraud



Claim Officer



Fraud Investigation Team



Regional Manager



Chief Risk Officer

Everyone receives alerts based on severity.

9. Advanced Real-Time Fraud Detection

The system should detect fraud **immediately**, not after hours or days.

Example

Customer submits claim

↓

AI checks instantly

↓

Within a second

↓

Claim is approved

or

Blocked

or

Sent for investigation

10. Adaptive Analytics

Adaptive means

The system changes automatically as fraud patterns change.

Example

Fraudsters start using fake mobile numbers.

AI notices.

AI changes its detection model automatically.

11. Explainable AI

Normally AI says

"Fraud"

But nobody knows why.

Explainable AI gives reasons.

Example

Fraud Score = 92%

Reasons

- ✓ Claim amount unusually high
- ✓ Hospital blacklisted
- ✓ Same bank account used before
- ✓ Customer claimed recently

This helps investigators trust the AI.

12. Fraud Investigation Workflow

The system should not stop at detection.

It should help investigators.

Example

AI detects fraud

↓

Case created

↓

Evidence attached

↓

Assign investigator

↓

Track investigation

↓

Close case

Everything happens inside the system.

13. Real-Time Fraud Decision Engine

This is the **brain** of the Fraud Management System.

Whenever a transaction happens, it decides instantly.

It can

Approve

Reject

Hold

Challenge

14. Inline Fraud Decisioning

Inline means

Checking fraud **while the transaction is happening**, before it is completed.

Example

Customer submits

₹25 lakh claim.

Before payment

AI checks.

Decision

Approve

Reject

Hold

Challenge

No waiting.

15. Real-Time Approve / Reject / Challenge

Possible decisions include:

- **Approve** – Transaction looks genuine.
- **Reject** – Clearly fraudulent.
- **Challenge** – Ask for extra verification (for example, additional documents or OTP).

16. Sub-Second Decision Latency

Latency means response time.

Sub-second means

Less than one second.

Example

Claim submitted

↓

AI checks

↓

0.3 seconds

↓

Decision returned

Very fast.

17. Synchronous Fraud APIs

API = connection between systems.

Synchronous means

The calling system waits immediately for the fraud decision.

Example

Claim System

↓

Calls Fraud API

↓

Waits

↓

Gets response

↓

Processes claim

18. Example Actions

The system should be able to

- Block a suspicious payment instantly.

- Alert the underwriting team before issuing a risky policy.
- Hold a claim until an investigator reviews it.

19. Streaming Fraud Decision Engine

Instead of checking only one event, the system continuously watches **live data streams**.

It monitors:

- Transactions
- Login activities
- Customer behavior
- Device information
- API requests

20. Event Correlation

AI connects related events.

Example

One customer

Logs in from Delhi

↓

After 2 minutes

Logs in from London

↓

Submits claim

AI links these events and marks them suspicious.

21. Complex Event Processing (CEP)

CEP means the system looks at a **sequence of events**, not just one event.

Example

Five wrong password attempts

↓

Successful login

↓

Large claim submitted

↓

Bank account changed

Individually these may seem normal, but together they indicate possible fraud.

Examples of CEP

Multiple failed logins followed by fund transfer

Someone tries many passwords, finally logs in, then immediately transfers money.

Very suspicious.

Geographically impossible access

Customer logs in from Bangalore.

Five minutes later

Another login from New York.

Impossible.

Likely account compromise.

Rapid policy creation followed by claim submission

Policy purchased today.

Claim submitted tomorrow.

High fraud probability.

CEP Features

The engine should support:

- **Rule chaining** – One rule triggers another.
- **Sliding windows** – Analyze events within a recent time window (for example, last 30 minutes).
- **Event aggregation** – Combine many small events into one analysis.
- **Temporal conditions** – Check the order and timing of events.

22. Real-Time Rules Engine

A configurable rules engine allows fraud experts to create or modify fraud rules without changing application code.

It should support:

- Dynamic rule creation.
- Rule prioritization.
- Threshold-based checks (for example, claim amount > ₹10 lakh).
- Context-aware risk policies.
- Business-user configurable workflows.

Additional Rules Engine Features

- **Hot deployment** – Publish new rules immediately without restarting the system.
- **Rule versioning** – Keep older versions for rollback and auditing.
- **Simulation/testing mode** – Test rules on historical data before activating them.
- **Rule auditability** – Record who created or modified a rule and when.

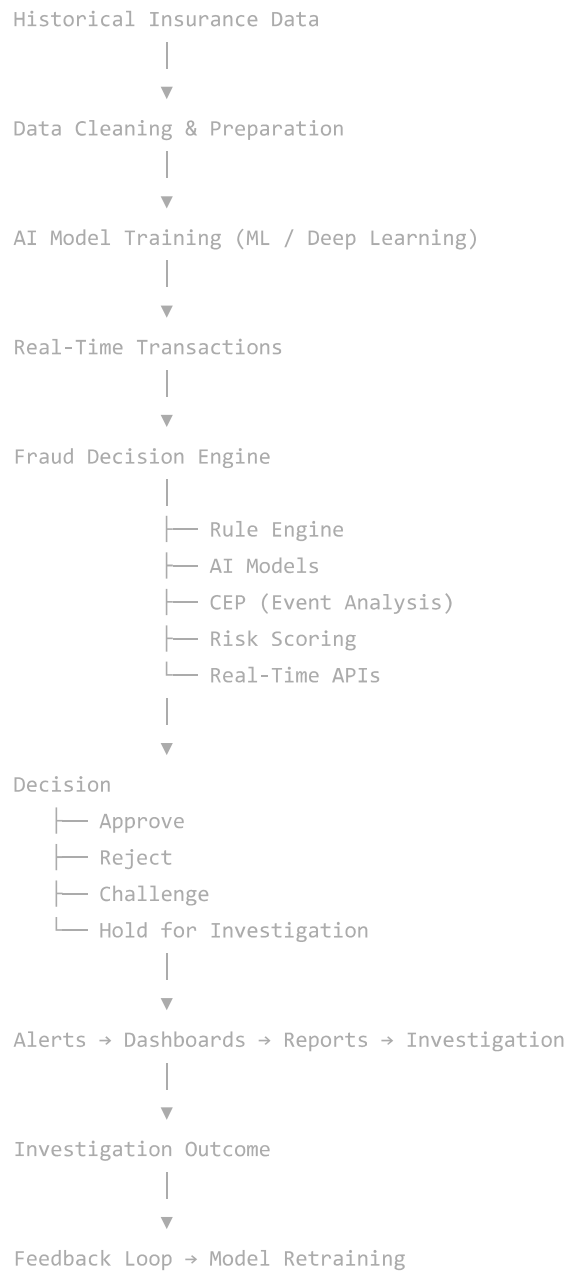
23. Real-Time Scoring APIs

Other insurance applications should be able to call the fraud engine through APIs to receive instant risk scores.

The APIs should provide:

- **Transaction risk score** – Is this transaction risky?
- **Customer risk score** – Is this customer high risk?
- **Behavioral anomaly score** – Is the customer's behavior unusual?
- **Device reputation score** – Has this device been involved in suspicious activities before?

Overall Flow of the Fraud Management System



In one sentence:

This RFP requires an enterprise-grade AI-based Fraud Management System that **learns from historical data, monitors every insurance transaction in real time, calculates fraud risk, explains its decisions, automates investigations and reporting, continuously improves through feedback, and integrates with other insurance systems using fast APIs.**



This section describes **how the AI models should continuously improve, remain accurate over time, defend against fraudsters trying to fool them, and comply with governance and regulatory requirements.**

Think of it like this:

The AI should **learn like an experienced fraud investigator**, improve every day, keep records of every change, and be able to explain every decision it makes.

1. Adaptive / Self-Learning Fraud Models

Simple Meaning

The AI should **not remain the same forever**.

It should continuously learn from new fraud cases and become smarter over time.

Why is this needed?

Fraudsters constantly invent new tricks.

If the AI only learns once, after a few months it becomes outdated.

Instead, it should keep updating itself.

Example

Initially, fraudsters submit fake hospital bills.

After some time, they start using AI-generated fake documents.

The system notices this new pattern and learns it automatically.

This is called a **self-learning model**.

2. Online Learning

Simple Meaning

Instead of retraining the AI from scratch every few months, it should **learn a little whenever new information becomes available**.

Imagine teaching a child.

✗ Old method

Every month you erase everything and teach the entire syllabus again.

✓ Online learning

Whenever the child learns one new thing, you simply add it to their knowledge.

The AI should work in the same way.

a) Incremental Learning

Meaning

The model should learn **small updates continuously**.

Example

Yesterday

100 fraud cases were confirmed.

Today

20 more fraud cases are confirmed.

Instead of retraining using all 120 cases from the beginning, the AI simply learns from the 20 new ones.

This saves time and computing resources.

b) Continuous Model Updates

Meaning

The AI should improve regularly.

Instead of updating once a year, it can update daily or weekly.

c) Near Real-Time Retraining

Meaning

When new fraud cases are identified, the AI should update **quickly**, not after months.

Example

Morning

A new fraud technique is discovered.

By afternoon

The AI has already learned it.

d) Feedback-Driven Learning

Meaning

The AI should learn from human investigators.

Example

AI predicts

"Fraud"

Investigator says

"No, this claim is genuine."

The AI remembers this correction.

Next time, it becomes more accurate.

The AI should learn from:

Newly confirmed fraud cases

Whenever investigators confirm that a case was fraudulent, the AI learns from it.

Analyst Feedback

Fraud analysts review suspicious claims.

Their decisions should improve the AI.

Evolving Fraud Patterns

Fraud methods keep changing.

The AI should continuously recognize these new patterns.

Without complete offline retraining

Meaning

The entire AI should **not need to be rebuilt from scratch** every time new data arrives.

It should simply update itself.

3. Concept Drift Detection

This is one of the most important AI concepts.

Simple Meaning

Sometimes the world changes.

The AI must detect that its **old knowledge is no longer accurate**.

This change is called **Concept Drift**.

Example

In 2024

Most fake claims came from fake hospitals.

In 2026

Fraudsters now use stolen identities instead.

The AI notices

"My old fraud patterns are no longer working."

That is concept drift.

The system should identify:

Changing Fraud Behaviors

Fraudsters invent new methods.

The AI should recognize this.

Degradation in Model Accuracy

Suppose earlier

AI accuracy = 97%

After six months

Accuracy falls to 82%.

The system should detect this drop automatically.

Evolving Transaction Patterns

Customer behavior changes over time.

The AI must distinguish between **normal business changes** and **actual fraud**.

The solution should:

Monitor Model Drift Metrics

Meaning

The system should continuously measure whether the AI is becoming less accurate.

Example metrics include:

- Accuracy
- Precision
- Recall
- False positives
- False negatives

If these worsen, it indicates drift.

Trigger Retraining Recommendations

When the model starts performing poorly, the system should recommend retraining.

Example

"Model accuracy has dropped from 97% to 89%. Retraining is recommended."

Adaptive Threshold Tuning

The fraud score threshold should adjust based on changing conditions.

Example

Previously

Fraud score > 70 = suspicious.

After fraud patterns change

The threshold might become 60 or 80.

The system should adjust this intelligently.

4. Adversarial Fraud Adaptation

Simple Meaning

Fraudsters also try to fool AI systems.

The AI must recognize these tricks and defend itself.

Fraud Pattern Mutation

Meaning

Fraudsters slightly change their methods to avoid detection.

Example

Old fraud

Claim amount = ₹10,00,000.

New fraud

Claim amount = ₹9,95,000.

The fraud looks slightly different, but the AI should still catch it.

Synthetic Identity Attacks

Meaning

Fraudsters create **fake identities** by combining real and fake information.

Example

Real Aadhaar number

-

Fake address

-

Fake phone number

The identity appears genuine but is fraudulent.

The AI should detect such cases.

Bot-Driven Behavioral Manipulation

Meaning

Automated software (bots) pretends to behave like real users.

Example

A bot creates thousands of fake insurance accounts.

The AI should identify these unusual patterns.

Model Evasion Attempts

Meaning

Fraudsters deliberately change their behavior to avoid being detected.

Example

If they know claims above ₹5 lakh are checked carefully, they submit many claims of ₹4.9 lakh instead.

The AI should recognize this tactic.

The system should provide:

Anomaly-Based Fallback Detection

If the normal fraud model is uncertain, the system should use **anomaly detection** as a backup.

Example

The AI has never seen this fraud pattern before.

Instead of ignoring it, anomaly detection flags it as unusual.

Adversarial Robustness Monitoring

The system should continuously check whether fraudsters are successfully fooling the AI.

If attacks increase, administrators should be alerted.

Adaptive Feature Engineering

Simple Meaning

The AI should automatically discover **new useful data points (features)** when fraud patterns change.

Example

Earlier, claim amount was important.

Now, login device type becomes more important.

The AI should adapt and use the new feature.

5. Model Risk Management (MRM)

This section is about **governing and controlling AI models**.

The AI should not be a "black box."

Everything should be documented and auditable.

6. Model Governance

Simple Meaning

The organization should know which AI models exist, who approved them, which version is running, and when they were changed.

Model Inventory Management

Maintain a list of all AI models.

Example

Model	Purpose
Fraud Model V1	Claim fraud detection
Fraud Model V2	Underwriting fraud
Behavioral Model	Login fraud detection

Model Version Control

Like software versions.

Example

- Version 1.0
- Version 1.1
- Version 2.0

If a new version performs poorly, the previous version can be restored.

Approval Workflows

A new model should **not go live automatically**.

Example workflow:

Developer → Data Scientist → Risk Team → Manager → Production.

Lifecycle Governance

The system should track the entire life of each model:

- Created
- Tested
- Approved
- Deployed
- Updated
- Retired

7. Model Validation

Before using a model, it must be tested thoroughly.

Bias Testing

The AI should treat similar cases fairly and not unfairly favor or disadvantage certain groups.

Example

Two customers with similar risk profiles should receive similar fraud assessments.

Performance Validation

The model should be evaluated for accuracy and reliability before deployment.

Example metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Challenger vs Champion Models

Simple Meaning

The current production model (Champion) competes against a new candidate model (Challenger).

Example

Champion accuracy = 95%

Challenger accuracy = 97%

If the challenger consistently performs better, it replaces the champion.

Independent Validation Workflows

Someone **other than the model developer** should independently verify that the model is correct and safe before deployment.

This reduces the risk of unnoticed mistakes.

8. Regulatory Traceability

Insurance regulators may ask:

"Why did the AI reject this claim?"

The system should be able to answer completely.

Model Lineage

Maintain the history of each model.

Example

Model V1 → V2 → V3

including what changed between versions.

Feature Lineage

Track which data fields (features) influenced the model.

Example

- Claim amount
- Customer age
- Hospital
- Device ID
- Previous claims

If a feature is removed or added, that history should also be recorded.

Decision Traceability

Every AI decision should be explainable.

Example

Claim rejected because:

- Fraud score = 94%
- Blacklisted hospital
- Duplicate invoice
- Device previously linked to fraud

This provides a clear audit trail.

Training Dataset History

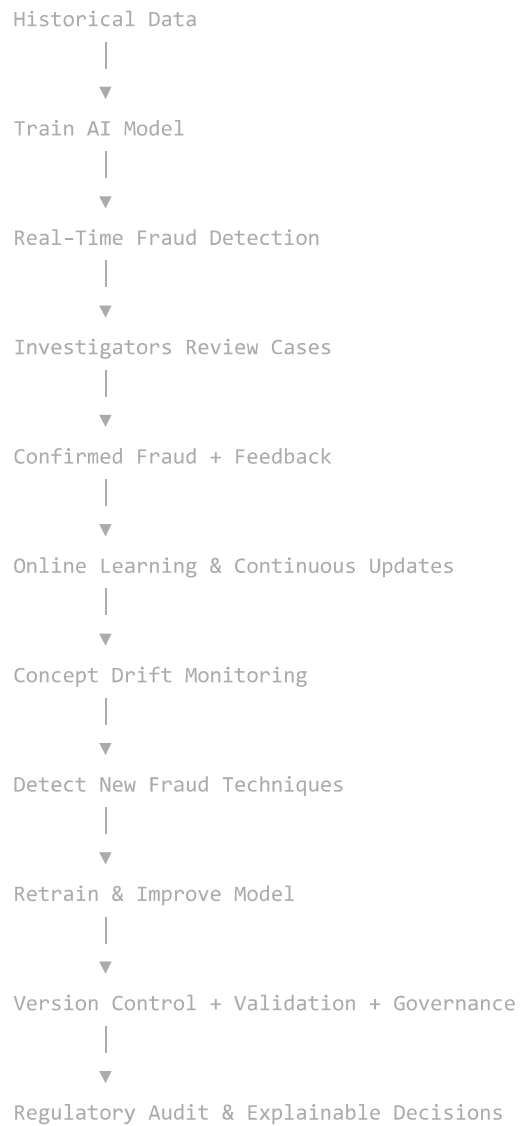
The system should record **exactly which data was used to train each model version**.

Example

Model Version	Training Data
V1	Data from 2023–2024
V2	Data from 2023–2025
V3	Data from 2023–2026

This is essential for audits, reproducibility, and regulatory compliance.

Overall Flow



In one sentence:

This section requires the Fraud Management System to have **self-learning AI models that continuously improve from new fraud cases and investigator feedback**, automatically detect when fraud patterns change, defend against attempts to fool the AI, and maintain complete governance, validation, version history, and audit trails so every AI decision is transparent and regulator-ready.



Explainable AI (XAI)

The solution shall provide Explainable AI (XAI) capabilities enabling transparent and auditable fraud decisions.

Explainable Alerts

Every fraud alert generated by AI/ML models shall provide:

- reason codes
- contributing risk factors
- confidence score
- feature importance indicators

Example Use case:

"Alert triggered due to unusual device behavior, abnormal transaction

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This is one of the **most important sections** of the RFP because it explains **how the AI should justify its decisions**, **how fraud investigations should be managed**, and **how the system should identify organized fraud rings by connecting related people and entities**.

PART 1: Explainable AI (XAI)

What is Explainable AI (XAI)?

Simple Meaning

Normally, AI gives an answer like:

✗ "This claim is fraudulent."

But nobody knows **why**.

Explainable AI (XAI) means the AI should also explain **why it reached that decision**.

Instead of just saying:

"Fraud detected"

it should say:

"Fraud detected because:

- **"The customer used a suspicious device."**
- **"The claim amount is unusually high."**
- **"The customer logged in from a different country."**
- **"Similar fraud was seen before."**

So investigators and auditors can understand the decision.

Explainable Alerts

Every fraud alert should include four things.

1. Reason Codes

Meaning

Short reasons explaining why the alert was generated.

Example:

- Duplicate bank account
- Blacklisted hospital
- Multiple recent claims
- Device mismatch
- Unusual login location

2. Contributing Risk Factors

These are the factors that increased the fraud risk.

Example

Claim Amount = ₹18 lakh

Hospital already under investigation

Customer submitted three claims in one month

All these increase the fraud score.

3. Confidence Score

This tells how confident the AI is.

Example

Fraud Score	Confidence
95%	Very High
80%	High
55%	Medium
20%	Low

Higher confidence means the AI is more certain.

4. Feature Importance Indicators

Feature means the information used by AI.

Feature importance tells which information influenced the decision the most.

Example

Feature	Importance
Device ID	45%
Hospital	30%
Claim Amount	15%
Customer Age	10%

Example

Instead of

"Fraud"

The system should show

Fraud Alert

Reasons:

- "Suspicious device"
- "Multiple claims in 7 days"
- "Login from unusual location"
- "Hospital already flagged"

Confidence = 96%

Analyst Explainability Dashboard

This is a dashboard used by fraud investigators.

Instead of reading technical AI results, they see visual explanations.

The dashboard should provide:

- Visual explanation charts
- Risk factors
- Fraud reasons
- Customer history
- Claim history
- Transaction details

SHAP / LIME Explainability

These are popular AI explainability techniques.

Simple explanation

Imagine AI is a teacher who gives marks.

SHAP and LIME explain:

"Why did the student receive 85 marks?"

Similarly,

they explain

"Why did AI say this claim is fraudulent?"

The RFP specifically requires support for **SHAP or LIME**, which are industry-standard methods for explaining AI model predictions.

Fraud Feature Attribution

This means identifying **which features contributed most** to the fraud decision.

Example

The fraud score was mainly influenced by:

- Device ID
- Claim Amount
- Hospital
- Bank Account

Transaction-Level Interpretability

Every individual claim or transaction should have its own explanation.

Not just overall model explanations.

Regulatory Explainability

Insurance companies are audited regularly.

The AI should explain decisions to:

- Auditors

- Regulators
- Compliance teams
- Courts (if required)

Example

Customer says

"Why was my claim rejected?"

The company should be able to answer with evidence.

PART 2: Fraud Case Management & Workflow Engine

Once fraud is detected, the system should help investigators manage the entire investigation.

Fraud Case Lifecycle

Think of it like a police case.

Every fraud case passes through stages.

Fraud Detected

↓

Case Created

↓

Assigned to Investigator

↓

Evidence Collected

↓

Decision Taken

↓

Case Closed

Case Creation

Whenever fraud is detected

A new investigation case should be created automatically.

Alert Triaging

Many alerts arrive every day.

The system should prioritize them.

Example

Critical Fraud → Investigate immediately

Medium Risk → Review later

Low Risk → Monitor

Investigation Assignment

Cases should automatically be assigned to investigators.

Example

Health insurance fraud → Health fraud team

Motor insurance fraud → Motor fraud team

Escalation Management

If the investigator doesn't respond within the required time, the case should automatically be escalated to a manager.

Closure Workflow

Once the investigation is complete, the case should be closed properly with findings and actions recorded.

Workflow Engine

A workflow defines the investigation process.

The RFP requires configurable workflows, meaning the insurance company can customize them.

SLA Tracking

SLA = Service Level Agreement.

Example

High-risk fraud case must be investigated within 24 hours.

The system should track these deadlines and raise alerts if they are missed.

Maker-Checker Approval

This is a four-eyes principle.

Example

Investigator (Maker)

↓

Manager (Checker)

↓

Final approval

One person cannot approve everything alone.

Investigator Collaboration

Multiple investigators should be able to work together on the same case, share notes, and update findings.

Investigation Console

This is the investigator's main workspace.

It should include:

- All fraud alerts
- Case details

- Customer information
- Claim history
- Evidence
- Investigation notes

Timeline Reconstruction

The system should reconstruct everything in chronological order.

Example

10:00 AM → Policy created

11:00 AM → Claim submitted

11:05 AM → Device changed

11:10 AM → Bank account updated

This helps investigators understand the sequence of events.

Linked Entity Visualization

The system should visually show how customers, hospitals, agents, and accounts are connected.

Evidence Management

All evidence should be stored in one place, such as:

- Documents
- Photos
- Emails

- Investigation notes
- CCTV (if available)

Alert Correlation

Instead of treating alerts separately, the system should combine related alerts.

Example

Three different alerts all relate to the same customer.

The system links them together.

Audit & Compliance

The system should maintain:

- An immutable audit trail (records cannot be altered without leaving evidence).
- Complete investigation history.
- Logs of every analyst action.
- Compliance reports for regulators.

PART 3: Graph Analytics for Fraud Ring Detection

This is one of the most advanced capabilities in the RFP.

What is Graph Analytics?

Instead of looking at one customer at a time, graph analytics studies **relationships between people, organizations, and other entities**.

Think of it like a social network.

Customer



Hospital



Bank Account



Agent



Garage



Another Customer

AI looks at all these connections.

Why?

Many frauds are not committed by one person.

They involve groups:

- Customers
- Hospitals
- Agents
- Garages

- Employees
- Vendors

Together, they form a **fraud ring**.

Graph Data Model

Everything becomes a **node**.

Examples of nodes:

- Customer
- Policy
- Claim
- Vehicle
- Mobile Number
- Email
- Bank Account
- Address
- Device ID
- IP Address
- Agent
- Hospital
- Garage
- Surveyor
- Employee
- Vendor

Relationships (Edges)

Nodes are connected by relationships such as:

- owns
- filed
- paid-to
- referred-by
- treated-at
- repaired-at
- shares-address
- shares-device
- shares-bank-account
- shares-phone
- approved-by

Example

```
Customer A
  |
Shares Bank Account
  |
Customer B
  |
Filed
  |
Claim X
  |
Treated At
  |
Hospital Y
  |
Also Treated
  |
Customer C
```

The graph shows hidden relationships that may indicate fraud.

Fraud Ring Detection

The system should identify suspicious patterns such as:

- Multiple customers using the same bank account, phone number, email, address, device, or IP address.
- Many claims linked to the same hospital, garage, surveyor, agent, or vendor.
- Closed-loop relationships (for example, the same agent, hospital, and customer repeatedly working together).
- One agent creating many policies followed by quick claims.
- One device submitting claims for many unrelated customers.
- Groups with unusually high claim activity.
- Suspicious links involving employees, vendors, and external parties.

Graph Algorithms

The graph engine should support techniques such as:

- **Community detection** – Find groups that are heavily connected.
- **Connected component analysis** – Identify separate connected networks.
- **Shortest path analysis** – Measure how closely two entities are related.
- **Link prediction** – Predict possible hidden relationships.
- **Centrality analysis** – Find the most influential or frequently connected entities.
- **PageRank/influence scoring** – Rank important nodes based on their connections.
- **Similarity scoring** – Compare entities for similar behavior.
- **Entity resolution** – Determine whether two records refer to the same person or organization.
- **Fraud ring risk scoring** – Assign a risk score to an entire network.

Entity Resolution

Sometimes the same person appears with slightly different information.

The system should recognize that they are actually the same entity.

It should compare:

- Similar names (fuzzy matching)
- Similar addresses
- Phone numbers
- Email IDs
- Device IDs
- PAN/Aadhaar (where legally permitted)
- Bank accounts
- Nearby geographic locations
- Similar-sounding names (phonetic matching)

This helps identify duplicate or disguised identities.

Investigator Visualization

Investigators should see an interactive graph showing:

- Connected customers
- Fraud rings
- Risk scores for each node and relationship
- Timeline of events
- Filters (claim type, date, location, hospital, garage, agent, etc.)
- Ability to click any node and view the complete customer, claim, or policy profile

Real-Time Graph Scoring

The graph should be checked during:

- Policy issuance
- Claim registration

- Claim approval
- Vendor onboarding
- Payment processing

Example

A new claim arrives.

Before approval, the system checks:

- Is the customer connected to a known fraudster?
- Is the device already linked to fraud?
- Is the hospital blacklisted?
- Is the bank account shared with previous fraud cases?

Only then does it decide whether to approve or investigate.

Alert Generation

The system should raise alerts when:

- A new entity joins an existing suspicious cluster.
- A claim is linked to known fraud entities.
- Too many entities share the same attributes (bank account, phone, device, etc.).
- A fraud ring suddenly grows.
- A highly connected (high-centrality) entity appears repeatedly.
- New unusual relationships emerge among multiple parties.

Explainability for Graph Alerts

Every graph-based alert should clearly explain:

- Why the alert was generated.
- Which entities are involved.

- The relationship path between them.
- How close they are to known fraudsters (graph distance).
- Which attributes are shared.
- How much each factor contributed to the risk score.

Example

Claim C123 is high risk because:

- “The claimant shares a device ID with four previously flagged claimants.”
- “The claim is connected within two relationship links (“2 hops”) to a blacklisted garage.”
- “The shared device and garage connections significantly increased the fraud risk score.”

Integration Requirements

The graph analytics module should work seamlessly with:

- Fraud Case Management System
- Claims Management System
- Policy Administration System
- Customer Master/CRM
- Payment System
- Vendor, Garage, and Hospital databases
- Data Lakehouse
- SIEM/SOC (Security Information and Event Management / Security Operations Center), where required
- Real-time fraud scoring APIs

This ensures that graph analytics uses data from all relevant systems and shares its results with them.

Governance & Audit

The system should maintain complete records of:

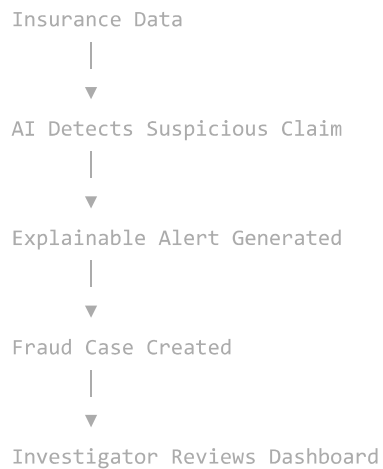
- Where graph data came from (data lineage).
- Which source system supplied each node and relationship.
- When nodes and relationships were created or updated.
- Every fraud alert generated.
- Every investigator action.
- Every model and rule version used.

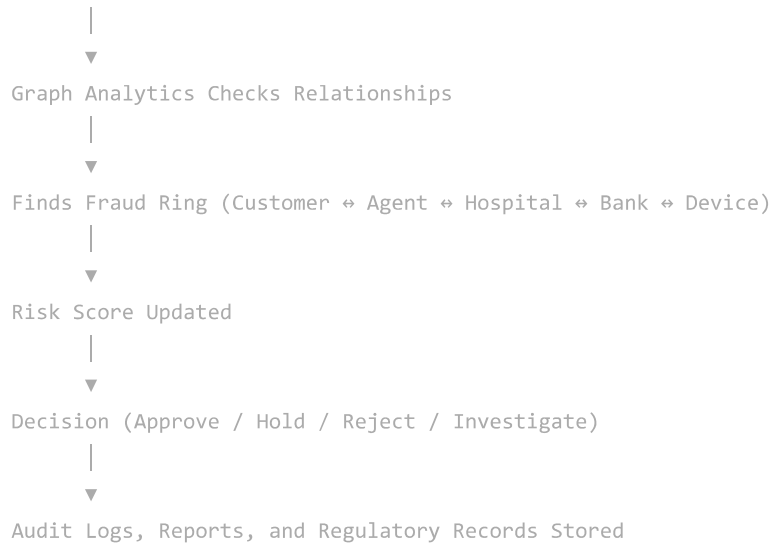
Performance Requirements

The graph engine should be capable of:

- Handling **very large networks** containing millions of entities and relationships.
- Looking up relationships with **very low latency**.
- Supporting both **batch updates** and **real-time streaming updates**.
- Efficiently traversing complex relationship networks.
- Providing **high availability** so the system remains operational.
- Enforcing **role-based access control (RBAC)** so users only see information they are authorized to access.

Overall Flow





In one sentence:

This section requires a Fraud Management System that **explains every AI decision, manages the entire fraud investigation process from alert to closure, and uses graph analytics to uncover hidden fraud rings by analyzing relationships among customers, claims, hospitals, agents, devices, bank accounts, and other entities, while maintaining full auditability, integration, and regulatory compliance.**



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Document

Explain in detail in simple words

This section describes the **most advanced capabilities** that the Fraud Management System (FMS) should provide. The goal is not only to detect fraud, but also to **explain why it is fraud, help investigators manage cases, and uncover organized fraud rings involving multiple people or organizations.**

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I'll explain each part in simple words with real insurance examples.

PART 1: Explainable AI (XAI)

What problem does Explainable AI solve?

Imagine an AI system simply says:

"Reject this claim."

The customer asks,

"Why?"

The manager asks,

"Why?"

The regulator asks,

"Why?"

If the AI cannot answer, nobody will trust it.

Therefore, the RFP requires **Explainable AI (XAI)**.

Instead of giving only a decision, the AI must also explain **how and why it reached that decision**.

Example without Explainable AI

Customer submits a ₹12 lakh health insurance claim.

AI says

✗ Fraud

No explanation.

The investigator has no idea why.

Example with Explainable AI

Customer submits ₹12 lakh claim.

AI says

Fraud Risk = 96%

Reasons:

- Hospital has high fraud history
- Claim submitted only 3 days after policy purchase
- Customer shares bank account with another fraud case
- Device was previously used in suspicious claims

Now everyone understands the decision.

Explainable Alerts

Whenever AI raises a fraud alert, it should include several details.

1. Reason Codes

Reason codes are **short explanations** describing why the alert was raised.

Example

Instead of

"Fraud"

The system should say

Reason Codes

- Duplicate Claim
- Shared Bank Account
- Suspicious Device
- Blacklisted Hospital
- High Claim Frequency

These are quick labels that investigators can understand immediately.

2. Contributing Risk Factors

A reason code says **what happened**.

A risk factor explains **what increased the fraud score**.

Imagine fraud scoring like exam marks.

Each suspicious activity adds marks.

Example

Activity	Risk Added
Policy bought yesterday	+20
Claim ₹15 lakh	+25
Same hospital flagged earlier	+30

Activity

Same mobile number in many claims

Risk Added

+20

Total Fraud Score = 95

These are contributing risk factors.

3. Confidence Score

Confidence tells **how sure the AI is**.

Example

Fraud Score = 92%

Confidence = 99%

Meaning

AI is almost certain.

Another example

Fraud Score = 60%

Confidence = 58%

Meaning

AI is uncertain.

Human investigation is required.

4. Feature Importance

AI uses many pieces of information called **features**.

Examples

- Claim amount
- Hospital
- Customer age
- Device ID
- IP address
- Previous claims
- Bank account
- Login location

Feature importance tells which features influenced the decision the most.

Example

Fraud Score = 91%

Reasons

Device ID → 45%

Hospital → 30%

Bank Account → 15%

Claim Amount → 10%

So the suspicious device contributed the most.

Example Explained

The RFP itself gives an example:

Alert triggered due to unusual device behavior, abnormal transaction velocity, and geolocation anomaly.

Let's understand this.

Unusual Device Behaviour

Normally

Customer always logs in using

Samsung Phone

Today

Claim submitted using

Unknown Laptop

Suspicious.

Transaction Velocity

Velocity means **speed or frequency**.

Normally

One claim every 2 years.

Suddenly

Three claims submitted within two hours.

Suspicious.

Geolocation Anomaly

Normally

Customer lives in Bangalore.

Today

Login from Russia.

One hour later

Claim submitted from Mumbai.

Impossible.

Suspicious.

Analyst Explainability Dashboard

This is the screen investigators will use.

Instead of showing only numbers, it should visually explain everything.

The dashboard should display:

- Fraud score
- Confidence
- Customer history
- Previous claims
- Hospital history
- Device history
- Timeline

- Reasons for fraud
- Related entities

It becomes a one-stop investigation screen.

SHAP and LIME

These are industry-standard AI explanation techniques.

Imagine a teacher giving 85 marks in an exam.

SHAP and LIME explain

Why not 90?

Why not 70?

Similarly,

They explain

Why fraud score became 91%.

Instead of saying

AI predicted fraud

they say

Because

Hospital contributed 40%

Device contributed 25%

Bank account contributed 20%

Location contributed 15%

This makes AI transparent.

Transaction-Level Interpretability

Every claim should have its own explanation.

Not

"The model generally detects fraud."

Instead

"This specific claim was flagged because..."

Every claim gets its own explanation.

Regulatory Explainability

Insurance companies are heavily regulated.

Suppose IRDAI asks

Why was this claim rejected?

The system should produce a report like

Fraud Score

Reasons

Evidence

Data used

Rules triggered

AI model version

Investigator comments

Everything should be available.

PART 2: Fraud Case Management

Detecting fraud is only the beginning.

Someone has to investigate.

The system should help manage the entire investigation.

Think of it like a police investigation system.

Fraud Case Lifecycle

Every case follows a sequence.

Fraud Detected

↓

Alert Generated

↓

Case Created

↓

Assigned to Investigator

↓

Evidence Collected

↓

Decision Taken

↓

Case Closed

Nothing should happen manually.

Case Creation

Suppose AI detects

95% fraud probability.

Immediately

The system creates

Case #F2026-4587

Automatically.

No employee needs to create it.

Alert Triaging

Suppose today

10,000 fraud alerts arrive.

Investigators cannot check all immediately.

The system prioritizes them.

Example

Critical

Claim ₹1 Crore

↓

Investigate immediately

High

₹20 lakh claim

↓

Investigate today

Medium

₹50,000 claim

↓

Review tomorrow

Investigation Assignment

Different investigators specialize in different frauds.

Motor Insurance Fraud

↓

Motor Team

Health Insurance Fraud

↓

Health Team

Marine Insurance Fraud

↓

Marine Team

The system automatically assigns cases.

Escalation Management

Suppose

Investigator doesn't respond for 24 hours.

Automatically

Manager gets notified.

If still delayed

Regional Head gets notified.

This is called escalation.

Closure Workflow

Once investigation finishes

The system records

Fraud Confirmed

or

Fraud Rejected

Recovery Amount

Penalty

Legal Action

Case Closed

Everything becomes part of history.

Workflow Engine

Every company has different procedures.

Therefore workflows should be configurable.

Example

OICL wants

AI



Fraud Team



Regional Manager



Legal Team



Closure

Another company may use a different sequence.

The workflow engine should allow this customization.

SLA Tracking

SLA means **Service Level Agreement**.

Example

High-risk claim

Must be investigated within

24 hours.

Medium-risk claim

Within

3 days.

The system tracks deadlines and alerts if they are missed.

Maker-Checker Approval

This prevents fraud by employees.

Example

Investigator recommends

Reject Claim.

Manager reviews the evidence.

Only after approval is the claim rejected.

One person cannot take the final decision alone.

Investigator Collaboration

Complex fraud cases often require several specialists.

The system should let them:

- Share notes.
- Upload evidence.
- Tag colleagues.
- Comment on findings.

Investigation Console

This is the investigator's main screen.

It should contain everything needed:

- Customer details
- Claim details
- Fraud score
- Evidence
- Timeline
- Previous claims
- Linked entities
- Investigation notes
- Final recommendation

The investigator should not have to switch between multiple systems.

Timeline Reconstruction

The system should recreate events in order.

Example

10:00 AM → Policy issued

10:20 AM → Address changed

10:40 AM → Bank account changed

11:00 AM → Claim submitted

11:05 AM → Large payment requested

Seeing the sequence helps investigators understand what happened.

Linked Entity Visualization

The system should visually show how entities are connected.

Example

Customer

↓

Policy

↓

Claim

↓

Hospital

↓

Bank Account

↓

Another Customer

Instead of reading many records, investigators immediately see hidden relationships.

Evidence Management

All investigation evidence should be stored in one place.

Examples:

- Claim documents
- Medical reports
- Photos
- Bills
- Emails
- Investigation notes
- Recorded interviews

Everything related to the case is available together.

Audit & Compliance

Every action should be recorded.

The system should know:

- Who viewed the case.
- Who changed information.
- Who approved or rejected it.
- When actions occurred.

These records should be immutable, meaning they cannot be secretly altered.

PART 3: Graph Analytics for Fraud Ring Detection

This is the most advanced requirement.

Why is Graph Analytics needed?

Looking at one claim at a time often misses organized fraud.

Fraud is frequently committed by groups working together.

For example:

- Customer
- Agent
- Hospital
- Garage
- Surveyor
- Employee

Each individual transaction may appear normal.

When you connect them, a fraud ring becomes visible.

What is a Graph?

A graph is a network.

Each object becomes a **node**.

Connections become **edges**.

Example



▼
Hospital Y

Now imagine millions of such connections.

The graph engine analyzes all of them.

Graph Nodes

Each real-world object becomes a node.

Examples include:

- Customer
- Policy
- Claim
- Vehicle
- Mobile Number
- Email ID
- Bank Account
- Address
- Device ID
- IP Address
- Agent
- Hospital
- Garage
- Surveyor
- Employee
- Vendor

Graph Relationships

The system connects nodes using relationships.

Examples:

Customer **owns** Policy.

Policy **has** Claim.

Claim **treated at** Hospital.

Claim **paid to** Bank Account.

Customer **shares device** with another customer.

These connections create a network.

Fraud Ring Example

Imagine five customers.

Normally they should be unrelated.

But graph analysis discovers:

- All use the same bank account.
- All use the same mobile number.
- All visited the same hospital.
- All policies were sold by the same agent.
- All claims were approved by the same employee.

Individually, each claim looks normal.

Together, they reveal an organized fraud ring.

Fraud Ring Detection Examples

The RFP expects the system to detect situations such as:

- Many customers sharing the same bank account, phone number, email, address, device, or IP address.
- Repeated claims involving the same hospital, garage, surveyor, agent, or vendor.
- Closed-loop relationships where the same customer, agent, hospital, and claim keep appearing together.
- An agent selling many policies that are followed by unusually quick claims.
- One device or IP address submitting claims for unrelated customers.
- Groups with unusually high claim activity.
- Suspicious collaborations involving employees, vendors, and external parties.

Graph Algorithms (Simple Meaning)

The graph engine should use algorithms to discover hidden relationships.

For example:

- **Community detection** groups closely connected entities that may form a fraud ring.
- **Shortest path analysis** measures how closely a customer is connected to a known fraudster.
- **Centrality analysis** finds the most influential or frequently connected entities, such as a hospital appearing in hundreds of suspicious claims.
- **Link prediction** predicts hidden relationships that may not yet be obvious.
- **Entity resolution** identifies when two records actually represent the same person despite slightly different details.

Real-Time Graph Scoring

When a new claim arrives, the graph should immediately answer questions like:

- Is this customer connected to any known fraudster?
- Is the bank account already linked to suspicious claims?
- Is the device shared with other unrelated customers?
- Is the hospital part of a high-risk cluster?

If the answer is yes, the claim receives a higher fraud risk score before processing.

Explainable Graph Alerts

The graph engine should not only say "High Risk." It should explain why.

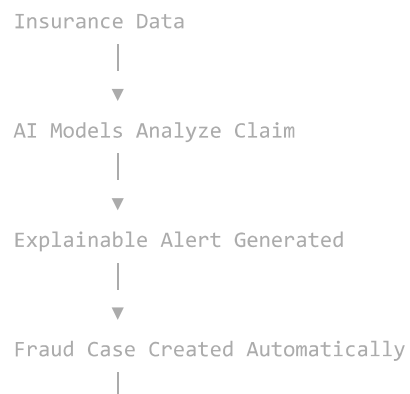
For example:

Claim C123 is high risk because:

- “The claimant shares a device ID with four previously flagged claimants.”
- “The claimant is connected within two relationship links ("2 hops") to a blacklisted garage.”
- “These shared relationships significantly increased the fraud risk score.”

This helps investigators quickly understand the network behind the alert.

Overall End-to-End Flow





In simple terms, this entire section asks the bidder to build an intelligent fraud platform that:

- Explains every AI decision so humans, auditors, and regulators understand it.
- Manages the complete investigation lifecycle from alert generation to case closure.
- Uses graph analytics to uncover organized fraud networks by connecting customers, claims, hospitals, agents, bank accounts, devices, and other entities that may appear unrelated when viewed individually.
- Maintains complete auditability, governance, and regulatory compliance throughout the process.

    Sources